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Routing algorithm selection for field coverage planning based on field shape and fleet size

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ABSTRACT

This project considered the field efficiency of routes created using two routing algorithms, a fast Clarke-Wright heuristic and a slower tabu search meta-heuristic, in 100 different field shapes with 1200 different field work scenarios for each algorithm. Different field work scenarios were generated by varying the number of simultaneously working vehicles (1, 2, 3, and 5) and using three path generation methods. After calculating the routes for all the scenarios using cloud computing and a high-performance compute cluster, the routes were evaluated to determine the field efficiency and field completion time. All tabu search results had field efficiencies above 0.65 and field completion times tightly clustered based on the number of simultaneously working vehicles. Many Clarke-Wright results were clustered with the tabu search results, but others had field efficiencies below 0.37 which indicated that the method failed to produce acceptable results. A logistic regression model was developed to identify the factors that caused unacceptable Clarke-Wright results. This enabled creation of a probability equation for predicting when the fast Clarke-Wright equation would produce acceptable results, and conversely, when it would be necessary to rely on the slower tabu search routing. Field shape complexity as measured by the isoperimetric quotient was the primary indicator that Clarke-Wright could fail to produce an acceptable result. The number of simultaneously operating vehicles was statistically insignificant for one, two or three vehicle fleets, but increasing to five vehicles did adversely affect the ability of the Clarke-Wright algorithm to provide acceptable routes.

1. Introduction

Field coverage planning is the process of breaking down the area of a field into paths that can be worked by agricultural field machinery, assigning paths to individual machines in a fleet, and creating an order in which the paths will be worked (Conesa-Muñoz et al., 2016; Hameed, 2014). This process generates a route that each piece of machinery is then able to follow as it carries out the field operation. Field coverage planning is important in improving the efficiency of traditional agricultural machinery operations and in enabling new autonomous systems (Bochtis et al., 2014). Current commercial systems are capable of performing almost all in-field navigation actions autonomously, but providing optimal entire field coverage plans built on this autonomous navigation capability is still an area of research (Baillie et al., 2018).

A major challenge of field coverage planning is that the problem is NP-Hard (Oksanen and Visala, 2009), which is a class of problems defined by computer scientists with the NP standing for non-deterministic polynomial time (Weisstein, 2018). Optimal solutions are therefore computationally intractable. Routing problems in general face this challenge, which has forced the development of algorithms that can produce reasonably optimal solutions in reasonable amounts of time where reasonable is defined by the specific problem under consideration (Toth and Vigo, 2002). Traditional heuristic methods were developed in previous decades when computer processing power was limited. More recent meta-heuristic algorithms, like genetic algorithms, simulated annealing and tabu search (Glover, 1989; Osman, 1993), are designed to perform a more thorough search of the possible solutions but are computationally more expensive. The nature of the problem and the design of the meta-heuristic determine both how long the calculation will take and the degree of optimization that is achieved (Osman, 1993).

Solution methods to field coverage planning are an active area of

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research for a variety of different types of operations. Studies have confirmed that route optimization can improve field operations when applying liquid fertilizer (Jensen et al., 2015), working flat fields of a variety of shapes (Jin and Tang, 2010; Oksanen and Visala, 2009) and when working fields with complex topography that places restrictions on certain travel paths (Hameed, 2014; Jin and Tang, 2011). Beyond improving basic route efficiency, field coverage planning has also been used to reduce soil compaction (Bochtis et al., 2012) and servicing times of support vehicles (Spekken and de Bruin, 2013). A significant challenge in field coverage planning of agricultural operations is their general unpredictability. Weather affects the areas of a field that are available for working on any given day (American Society of Agricultural and Biological Engineers, 2011a). Machinery breakdowns and worker obtainability can change the number of machines that can be used in a given field operation (American Society of Agricultural and Biological Engineers, 2011b). A computationally-expensive meta-heuristic may produce a better solution, but it may require more expensive computer hardware, offline processing days in advance, or purchasing significant amounts of cloud-based computer resources.

While metaheuristic methods generally identify more optimal solutions than heuristic methods (Toth and Vigo, 2002), the differences appear to depend significantly on the complexity of the operation (Seyyedhasani and Dvorak, 2017). There is less difference in a simple rectangular field worked with one machine than in a field with a complicated shape worked simultaneously by 10 machines. As engineers begin to apply field coverage planning algorithms in agricultural machinery, an important consideration is when to switch between a light-weight algorithm to a computationally expensive one. This is true whether it is applied in the navigation, guidance and fleet telematics software built into modern manned machinery or the anticipated future fleets of autonomous robotic platforms (Blackmore et al., 2002; Blender et al., 2016; Pitla et al., 2010; Shearer et al., 2010).

Although producers may prefer meta-heuristics since they tend to deliver more efficient solutions, a pre-computed route with the exact same fleet and field composition or enough computational power to provide a meta-heuristic-based solution in a reasonable amount of time may not always be available during actual field operations. In this project, we investigated the degree of route optimization provided by a fast heuristic and a slower meta-heuristic method used for the path assignment and routing portions of field coverage planning under a variety of different levels of complexity. The heuristic and meta-heuristic methods were a modified version of the Clarke-Wright savings algorithm and tabu search, respectively. The degree of route optimization was measured by overall field efficiency, which is the ratio between the productivity of a machine under field conditions and the theoretical maximum productivity (American Society of Agricultural and Biological Engineers, 2011b, c). Turning time and other non-productive times are a major component in field efficiency (American Society of Agricultural and Biological Engineers, 2011b). Complexity was generated by different field shapes, numbers of vehicles and the method used to break the fields into paths. The goal was to develop a statistical model that enabled predicting when the fast heuristic method would perform acceptably and similarly to the slower and more complex meta-heuristic method.

2. Materials and methods

2.1. Description of operation to be simulated

Agricultural field operations are highly variable with different constraints and operating methods, and we had to select a single type of field operation for consistency. For this study, the field operation was conducted by homogeneous machines without support vehicles, refills or unloading and using the headland pattern. This pattern consisted of paths around the boundary of the field to create headlands with internal AB-lines to complete field coverage. The boundaries were continuous straight paths, which Graf Plessen (2018) called the ABp field pattern. This covers a wide variety of potential field work, and some examples include material-neutral operations like tillage, cultivation and mowing, and fertilizer or spray applications where refills in the field are not necessary. The field work model also forced vehicles to travel on the headlands when moving between paths. In other words, vehicles were not permitted to travel across the regular field paths or outside field boundaries, which is an important concern with non-convex field shapes. Certain tillage operations may not normally have this headland-only travel restriction, but planting and especially weed-control cultivation and spraying in standing row crops would require it.

Since a primary factor of interest was field shape, all fields were scaled to the same area and implement widths were considered identical. Studies have confirmed that large fields are more efficient than smaller fields (Renoll, 1970), and that smaller implement widths provide higher field efficiency than larger ones (Condra et al., 2017). Field structure also plays a role as long-row fields are more efficient than short-row fields (Renoll, 1970).

Based on the previously mentioned findings, a relationship would be expected between field efficiency and the path width to area ratio with interactions with field shape. In general, a lower path width to area ratio indicates more passes will be required to completely cover a field. More passes would require more headland turns and more non-productive time. Other studies have illustrated the critical role that headland turning plays in optimizing field efficiency (Paraforos et al., 2018). To isolate field shape from field size, the path width to area ratio was held constant in this study.

2.2. Performing the simulations

The boundaries of 100 fields were digitized. For each field, compactness of field shape was assessed using the isoperimetric quotient:

$$IQ = \frac{4\pi A}{P^2} \tag{1}$$

where

IQ is the isoperimetric quotient,
A is area of the field, and
P is the perimeter of the field.

The isoperimetric quotient has been used to assess field shape by other researchers (Machl et al., 2013; Zandonadi, 2012), and agricultural fields can be expected to range from 0.1 (highly irregular) to 1 (a perfect circle). For reference, a square has an isoperimetric quotient of 0.79. Fields were selected from different regions with a focus on capturing a variety of shapes from the oddly shaped fields associated with Kentucky and the eastern United States to the geometric shapes found in flat areas of western Kansas and the Midwest (Fig. 1).

The goal was to record a range of isoperimetric quotients to represent the full range of field shapes in which agricultural vehicles could be expected to operate. The minimum and maximum isoperimetric quotients in this set of fields were 0.13 and 0.99 (Fig. 2). Since a primary factor of interest was field shape, all fields were scaled to an identical area of 1 ha, and path width (i.e. implement width) was set to 2 m. (see Fig. 3).

Path lists were generated for each field using 3 techniques:

(1). All paths parallel with the longest field edge.
(2). Parallel paths with the direction selected for minimum double coverage (Zandonadi et al., 2011).
(3). Section decomposition using the method of Jin and Tang (2010) with direction set to minimize double coverage.

While these path generation methods do not completely capture the wide variety of methods used and desired by farmers (Mederle and

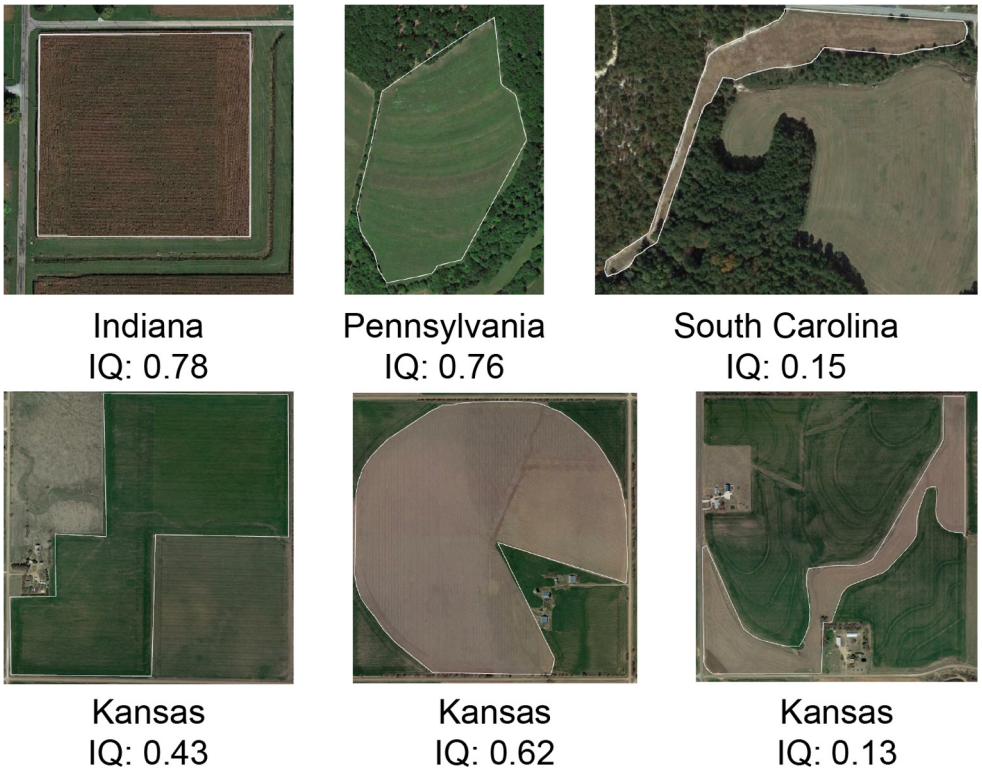


Fig. 1. Some of the field shapes used in this study. Fields on the left would be relatively simple to route with increasing difficulty toward those on the right.

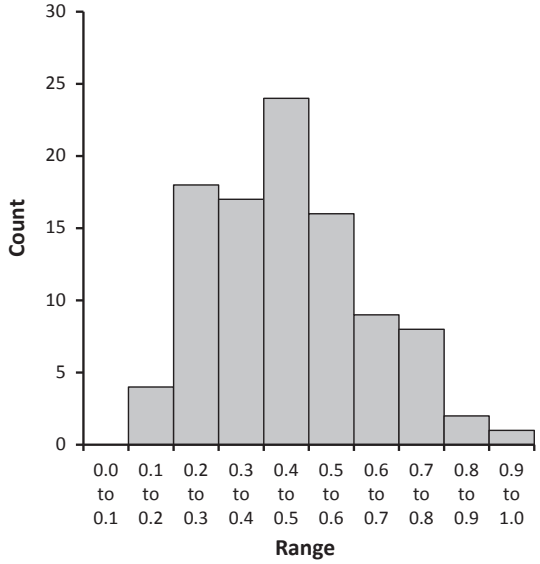


Fig. 2. Histogram of field isoperimetric quotients used in the simulations.

Bernhardt, 2017), following these three methods created fields with a variety of common field path structures for our routing algorithm testing. This created 300 path lists that represented the fields and provided complete area coverage. For each path list, scenarios were created using 1, 2, 3, and 5 simultaneously working vehicles. Following these steps produced 1200 scenarios for testing. For each scenario, solutions were generated using both the modified Clark-Wright and tabu search methods as described in Seyyedhasani and Dvorak (2017).

The calculations for performing section decomposition and the tabu search path optimization required considerable computational resources. With section decomposition following Jin and Tang (2010), each possible division of the field needed to be evaluated to determine if it produced an improved set of routes. Routes within each section were created and evaluated using the minimal double coverage area technique from Zandonadi et al. (2011). The mean and median number of sides per field in the sample set was 57 and 47, respectively, so there were many possible field divisions to evaluate for each field. Given the computational requirements, the section decomposition method was only used to generate one division rather than processing each section iteratively until the best set of sections had been created. This limitation enabled the section decomposition process for each field to be completed within at most several hours on a desktop running Intel i7-class processors. As a meta-heuristic, tabu search could likewise require up to several hours of processing for each scenario. To perform the calculations necessary for all the scenarios, processing for the 1,200 scenarios was performed on Amazon Web Services, Microsoft Azure and the University of Kentucky's High-Performance Compute Cluster.

2.3. Statistical analysis methods

Statistical analysis was conducted to determine the factors that significantly affected the field work parameters. The Clarke-Wright method is considerably quicker to calculate so the statistical analysis focused on identifying the characteristics under which it was as good as tabu search in route generation. To model the probability of an acceptable field efficiency for Clarke-Wright, a logistic regression was run on the binary indicator variable of acceptable field efficiency on the independent variables of the number of vehicles, the field IQ, and the path generation method. The best model with the combination of those variables, including interactions and polynomials, was determined by the smallest Akaike's Information Criteria (AIC) statistic. All analyses were run in SAS software, Version 9.4 of the SAS System for Windows in Proc Genmod. Statistical plots were constructed in ggplot2 (Wickham, 2009).

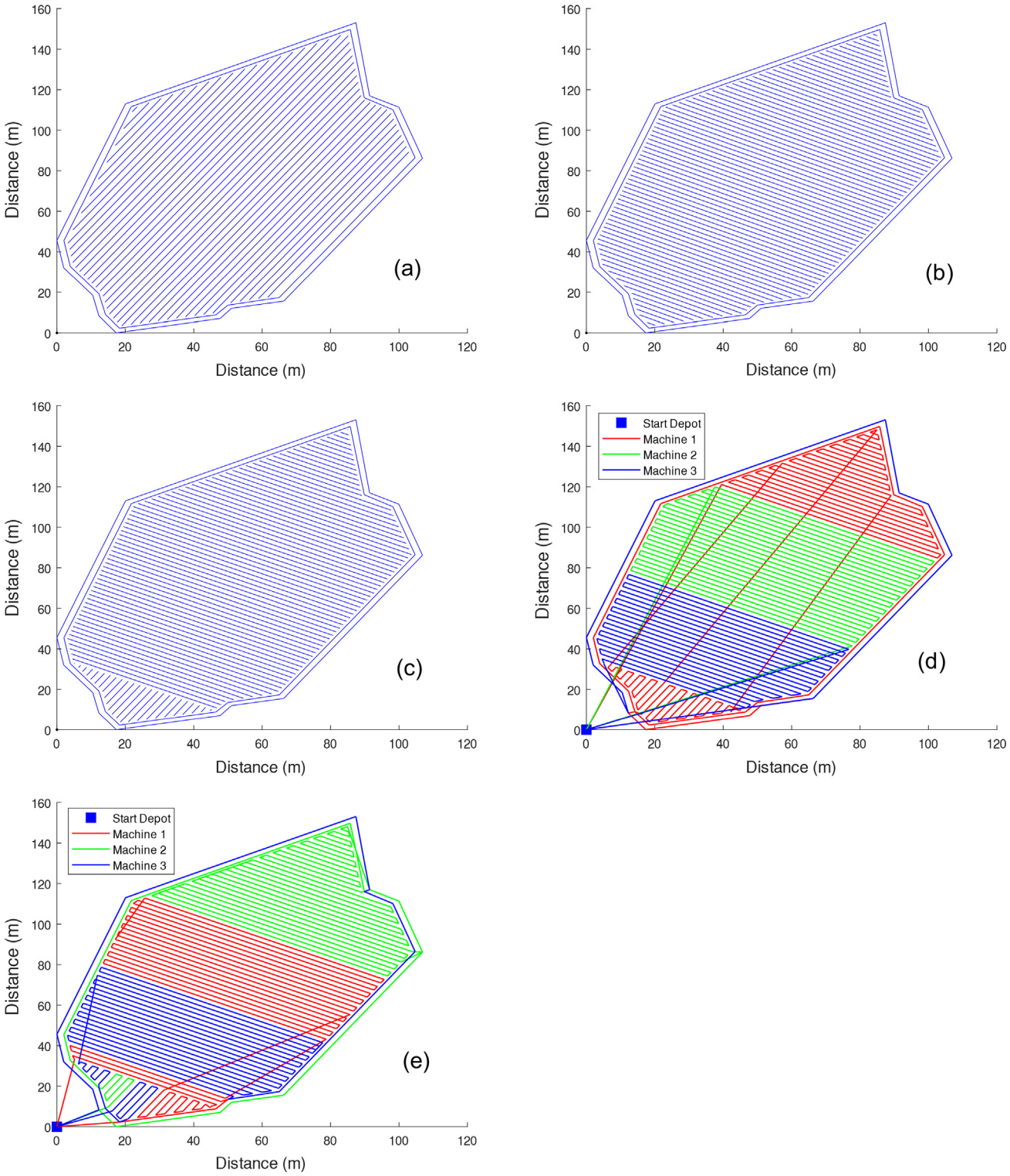


Fig. 3. Examples with the Pennsylvania field of Fig. 1 of path generation with (a) paths parallel with longest edge, (b) path direction to minimize double coverage, and (c) section decomposition and solutions using (d) Clark-Wright and (e) tabu search.

3. Results

When looking at the distribution of the field efficiencies from the simulations (Fig. 4), there is a clear separation of results between 0.4 and 0.7. Most field efficiencies were above 0.7, but another large portion ended up below 0.4 with no simulations resulting in field efficiencies between 0.4 and 0.7. In an operation with many stops, material transfers or other actions, field efficiencies as low as 0.5 may be expected; however, these simulations were based on a material-neutral operation where field efficiencies should be between 0.7 and 0.9 (American Society of Agricultural and Biological Engineers, 2011a). Therefore, it is safe to classify routes with field efficiencies below 0.4 as unacceptable.

A log-log graph of field efficiency to field completion time illustrates the separation that exists in field efficiencies (Fig. 5). The upper left-hand corner represents high field efficiencies (those above 0.7) and low

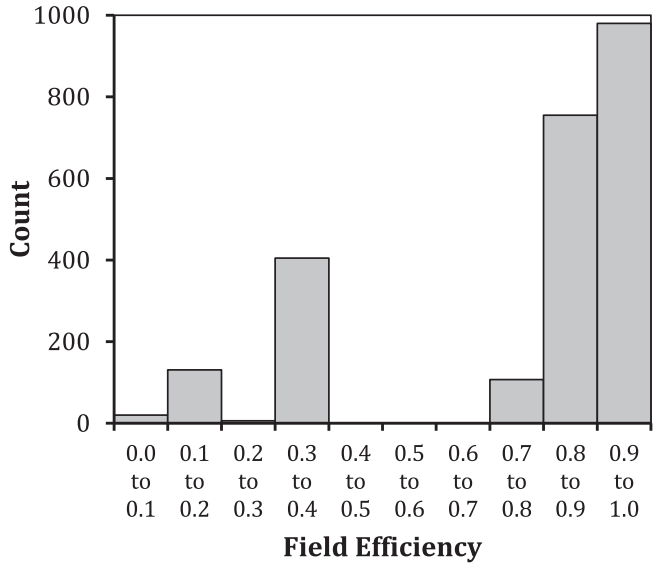


Fig. 4. Histogram of field efficiencies of the simulated routes.

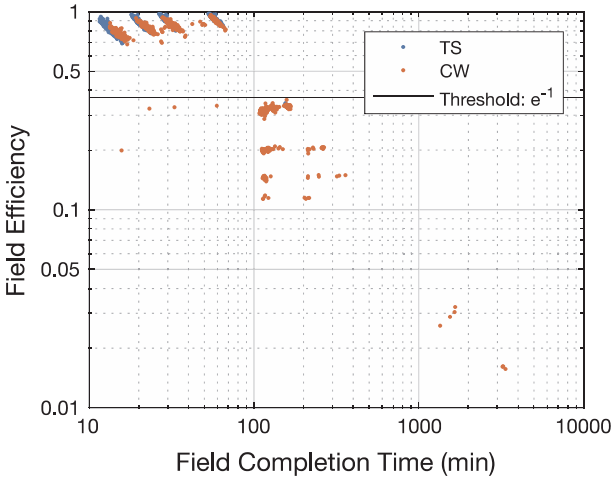


Fig. 5. A logarithmic plot of field completion time, field efficiency by the solution techniques (CW: Clarke-Wright, TS: Tabu Search) for the 2400 routing simulations. The cutoff threshold between acceptable and unacceptable field efficiencies (ln(F.E.) = −1 or e^{-1} = 0.37) is also shown.

field completion times. These represent acceptable routing solutions. Almost all tabu search results are located in this upper left-hand corner, while Clarke-Wright results are found at a variety of field efficiency levels. The unacceptable results can be extremely unacceptable which necessitated the use of a log-log graph to display all the results. A cutoff threshold between acceptable and unacceptable results was established at a field efficiency of 0.37, which is $\ln(field\ \ efficiency) = -1$ using the log representation of the figure. No tabu search results fall below this cutoff threshold.

It is also evident that there are four clusters of tabu search results. These correspond to various numbers of vehicles in the fleet (Fig. 6). The Clarke-Wright results (represented by a single point) often cluster directly with the tabu search results (represented by the +). This is especially true for the cluster representing a one-vehicle fleet. With higher numbers of vehicles in the fleet, the Clarke-Wright results tend to cluster at slightly higher completion times and at lower field efficiencies than tabu search. While Clarke-Wright appears to produce slightly worse results as the number of vehicles increases, these routes would still be within the range of acceptable routes.

Given the drastic difference between acceptable and unacceptable

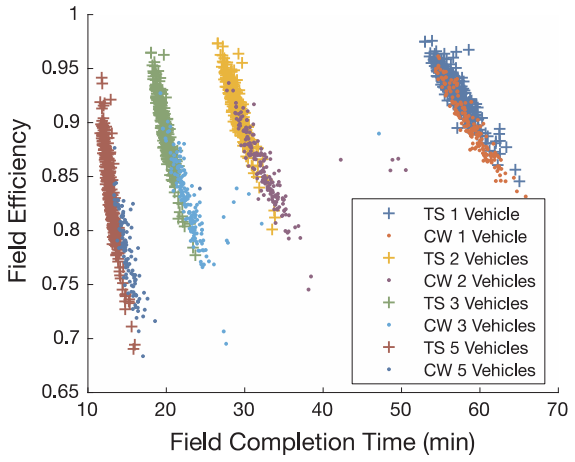


Fig. 6. A plot showing a subset of the points from Fig. 5 of the simulation results with acceptable field efficiency. The results are clustered by number of vehicles in the fleet.

results, the next step was to identify an equation to predict when Clarke-Wright would provide an acceptable solution (with acceptable classified as above the field efficiency cutoff threshold of 0.37). The logistic regression model provided the following equation for determining the probability of an acceptable field efficiency when using Clarke-Wright:

$$p = logit^{-1}(2.03 + 9.10 \times IQ - 7.06 \times IQ^2 - 0.87 \times FV) \quad (2)$$

where

p is the probability of an acceptable field efficiency,
IQ is the iso-parametric quotient,
FV is an indicator variable for five vehicles,
And $logit^{-1}$ is the inverse logit transformation, $logit^{-1}(x) = \frac{e^x}{1+e^x}$.

The Hosmer and Lemeshow Test did not indicate evidence of a lack of fit, with H = 11.2646 and an associated p-value = 0.1872. Moreover, the Area Under the (Reciever operating characteristic or ROC) Curve (AUC) is 0.6515, with confidence limits of (0.6207, 0.6824), which suggests the model is superior to random guessing.

This equation includes an indicator variable for the presence of five vehicles rather than one, two or three. The importance of this is apparent when the statistical model is used to plot probabilities of acceptable field efficiencies taking into consideration the isoperimetric quotient and the number of vehicles (Fig. 7). The probabilities for one, two and three vehicles are very close for most isoperimetric quotients but clearly different with five vehicles.

The equation for predicting an acceptable field efficiency with Clarke-Wright is a second-order polynomial of IQ. For one, two and three vehicle fleets, the peak probability occurs at an IQ of between 0.6 and 0.7, and probability decreases at IQs higher than this. At first glance, this may seem odd, but the travel pattern used for path generation was the common headland pattern, which is most efficient in rectangular and square fields. While the IQ of a perfect square is 0.79, rectangles have lower IQ, and small non-convex inclusions in square field boundaries, which are generally insignificant for route efficiency, similarly lower the IQ. These factors contribute to the peaks in the probability equations between 0.6 and 0.7. Although this work only used a consistent headland work pattern in all of the fields, a future work opportunity would be to investigate modifying the basic path generation method to utilize IQ and produce circular routes instead of headland routes above a certain IQ threshold.

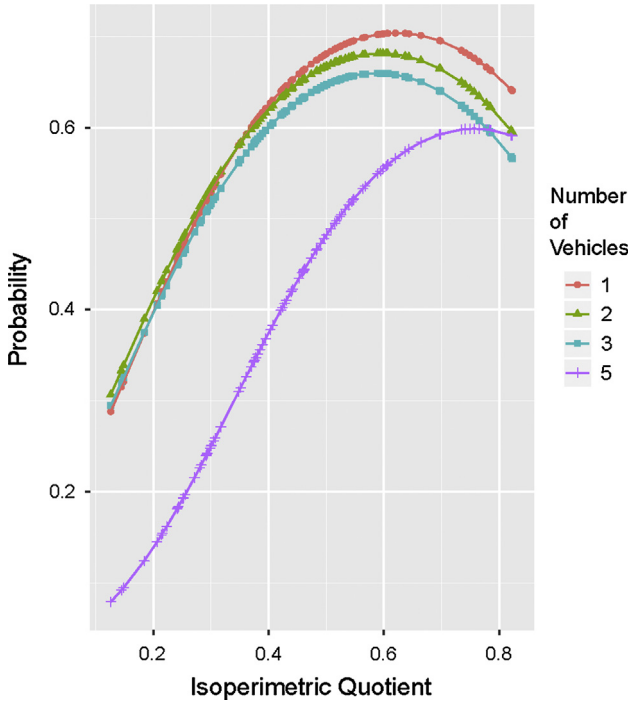


Fig. 7. Probability that the field efficiency will be greater than e^{-1}.

4. Discussion

The primary outcome of this project is an equation to predict when a simple Clarke-Wright route optimization should provide an acceptable route. In highly time-constrained situations, it could determine if automatic route optimization was even feasible as Clarke-Wright is the only choice in these cases.

It is expected that farm managers would choose to use more complicated meta-heuristic routing optimization algorithms, like tabu search, when time is not critical. In many cases, producers know the shape of their fields and the number of vehicles (tractors + implements) they have for a given type of operation. Even if managers are unwilling to dedicate their personal computers to route calculations for the significant time that can be required by meta-heuristic algorithms, a cloud service as used in this project could complete the calculation for a very nominal cost. Therefore, many field operations would be performed just following the pre-determined routes that have been calculated in the off-season.

However, farming operations do not always happen according to plan. Vehicles break down. Employees are sick and do not show up on time. Sometimes even the shape of the field changes because of weather conditions or crop buffer requirements. When these events happen, it will be possible to use a probability equation like the one created in this work to determine when it is likely acceptable to rely on Clarke-Wright results and when it would be necessary to wait for more advanced routing algorithms to create routes.

These results also affect dynamic route updating as dynamic updates are considered when field and equipment conditions have caused deviations from the initial solutions. These deviations would cause a loss of efficiency in the original routes, but Clarke-Wright is the most reasonable method for dynamic updates. If Clarke-Wright is expected to also yield a good set of routes given the number of vehicles and field configuration, it could be used to update routes. However, if the probability is low, the less efficient original routes could be maintained.

Even though this study included nearly 2400 simulations in 100 different field shapes and required thousands of hours of computer time, it must only be considered an initial analysis of this problem. Some researchers have proposed agricultural vehicles fleets composed of many more than five vehicles (Blender et al., 2016). Our work indicates that the probability of a good route with Clarke-Wright falls with five vehicles. It is, therefore, reasonable to assume that it could fall further with larger numbers of vehicles.

We have also isolated and analyzed the effect of field shape independent of field size. As other research has illustrated, field size has an effect on field efficiency by changing the headland area and adjusting the number of paths required to complete the field (Paraforos et al., 2018). It is possible that a metaheuristic like tabu search rather than a Clarke-Wright heuristic could minimize the loss of efficiency that happens when large implements are utilized in working small fields. This is area of future investigation.

Another limitation of this work is that the given probability equation provided is only valid for the exact algorithms used. Route optimization algorithms for agricultural vehicles is a very active area of research. Researchers are proposing new and more efficient methods to convert the field coverage problem into a Travelling Salesperson Problem or a Vehicle Routing Problem and are investigating new metaheuristics to provide optimized routes more efficiently (Conesa-Muñoz et al., 2016; Conesa-Munoz et al., 2012). Altering the field representation or improving the way the Clarke-Wright algorithm is implemented would adjust the probabilities of obtaining an acceptable routing. On the other hand, field variability and equipment issues will always be present, and agricultural operations will continue be performed in rural, distributed locations where instant access to high powered cloud-compute resources may not be guaranteed. There will also continue to be differences in computational requirements and route efficiencies between these newly published algorithms. As automatic route creation begins to appear in commercial equipment, it will be important to recreate the probability equation using the exact routing algorithms as implemented in the navigation controllers. Therefore, development of a commercial product based on this research will require repeating several steps to create a new probability equation. However, this work does illustrate the possibility of applying a fast algorithm and a probability equation to determine when that is an acceptable option.

5. Conclusion

The field efficiencies for all routes generated using tabu search were above 0.65. Most routes generated using Clarke-Wright were similar to the results from tabu search, but many others had unacceptable field efficiencies below 0.4. A probability equation was developed using a logistic model of the results to predict when Clarke-Wright would provide acceptable field efficiencies. The primary factor in this equation was the isoperimetric quotient, and complex field shapes with lower isoperimetric quotients were less likely to provide acceptable results with Clarke-Wright. The number of vehicles working the field was insignificant with one, two or three vehicles, but using five vehicles resulted in a lower probability of achieving an acceptable solution with Clarke-Wright.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compag.2018.12.002.

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